

How to treat autism in children?

Evidence searched on 2026-10-07 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **exercise interventions (e.g., dance, structured walking) to improve motor skills** in children with autism spectrum disorder (ASD), with dose-response effects reported for overall and gross motor function. **No direct evidence** was found for treatments targeting core ASD symptoms (e.g., social communication, behavior) or feeding difficulties in this population.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills. [1]

Immediate context Motor skill interventions show evidence for functional improvements, but **no direct evidence** was found for treatments addressing social communication, emotion regulation, or behavioral symptoms in ASD. Feeding difficulties were addressed in a different population (indirect evidence). Ongoing trials explore water competency and social responsiveness but have not yet reported results.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Irritability Associated with Autistic Disorder"	DailyMed
Risperidone	"Treatment of irritability associated with autistic disorder"	DailyMed
NT1164 (latest, investigational)	Being tested in a registered Phase 3 trial, Not yet recruiting; not an approved treatment	NCT07257939

Latest findings

- **2026-12: Case study**, Dual-target deep brain stimulation (DBS) in a 12-month-old child with severe autism showed sustained reductions in core symptoms (e.g., ATEC scores decreased by 81.3% at 12 months) [6].
- **2026-12: Systematic review**, Behavioral sleep interventions improved sleep outcomes in autistic children, with 39 studies included since 2002 [2].
- **2026-09-22: Case report**, Crohn's disease in an adolescent with autism was misdiagnosed due to communication barriers, highlighting the need for gastrointestinal screening [7].
- **2026-09-18: Meta-analysis**, Exercise interventions improved motor skills in children with ASD, with dose-response effects for gross and overall motor skills [1].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Exercise interventions	Evidence-supported but limited	Exercise significantly improved gross, fine, and overall motor skills in children with ASD, with dose-response effects.	[1]
Behavioral sleep interventions	Evidence-supported but limited	Systematic review of 39 studies (2002–2026) found behavioral sleep interventions effective for autistic children under 18.	[2]

Approach	Evidence level	What the evidence shows	Reference
Modified ketogenic diet	Emerging	Short-term reductions in ASD symptom severity (ABC/CARS scores) in children with ASD, combined with rehabilitation.	[3]
Caregiver-assisted NDBI	Emerging	Pilot RCT showed feasibility of group-based naturalistic developmental behavioral interventions for preschoolers.	[4]
Extended reality (XR) interventions	Emerging	Systematic review identified 113 XR studies (2020–2026), with VR/AR showing potential for training, therapy, and assessment.	[5]
Deep brain stimulation (DBS)	Experimental or preliminary	Single 12-month case study reported sustained reductions in core ASD symptoms (e.g., 81.3% improvement in ATEC scores).	[6]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Aripiprazole	Approved (FDA-labelled for this condition)	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder"	Route: Oral. Pediatric Patients (6 to 17 years) The recommended dosage range for the treatment of pediatric patients with irritability associated with autistic disorder is 5 to 15 mg/day. Dosing should be initiated at 2 mg/day. The dose should be increased to 5 mg/day, with subsequent increases to 10 or 15 mg/day if needed. Dose adjustments of up to 5 mg/day should occur gradually, at intervals of no less than one week.	FDA-approved for the use shown; label effective 2026-03-05; original approval date not retrieved	FDA label. DailyMed

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Risperidone	Approved (FDA-labelled for this condition)	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase the dose to 0.5 mg per day for children under 20 kg or 1.0 mg per day for those 20 kg or heavier, and maintain this dose for no less than 14 days. If needed, raise the dose by 0.25 mg per day for children under 20 kg or 0.5 mg per day for those 20 kg or more, at intervals of two weeks or more. The usual dose range is 0.5 mg to 3 mg per day.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed

Other drugs: used for symptoms, or still being tested (not established treatments of this condition)

Off-label: a retrieved review or page describes use for specific symptoms or co-occurring conditions; the drug has no FDA-labelled indication for this condition. Emerging: a registered late-phase trial; investigational, never an established treatment.

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Melatonin	Off-label (described as used in practice)	sleep problems in ASD	Not stated in the retrieved source for this use	No FDA-labelled indication for this condition in the retrieved labels	"Melatonin is commonly used to address sleep problems in ASD and appears to be effective." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
Quetiapine	Off-label (described as used in practice)	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Not stated in the retrieved source for this use	No FDA-labelled indication for this condition in the retrieved labels	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
Ziprasidone	Off-label (described as used in practice)	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Not stated in the retrieved source for this use	No FDA-labelled indication for this condition in the retrieved labels	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
Paliperidone	Off-label (described as used in practice)	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Not stated in the retrieved source for this use	No FDA-labelled indication for this condition in the retrieved labels	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology

Drug	Category	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
buspirone	Emerging: investigational, Phase 4	Registered conditions: Anxiety; Autism Spectrum Disorder. Ages: from 7 Years, up to 17 Years, child	Investigational: trial-specific dosing is in the registry record	Not approved. Recruiting, last update 2026-09-17	NCT07439042 . Primary outcome: Mean 16-Week Change in Pediatric Anxiety Rating Scale (PARS) 5-Item Total Score. A registry entry does not show that the drug works.
NTI164	Emerging: investigational, Phase 3	Registered conditions: Autism; Autism Spectrum Disorder; Autism Disorder. Ages: from 6 Years, up to 21 Years, child/adult	Investigational: trial-specific dosing is in the registry record	Not approved. Not yet recruiting, last update 2026-09-30	NCT07257939 . Primary outcome: Vineland Adaptive Behaviour Scales, 3rd Edition (Vineland-3). A registry entry does not show that the drug works.
KarXT	Emerging: investigational, Phase 3	Registered conditions: Irritability Associated With Autism Spectrum Disorder. Ages: from 5 Years, up to 17 Years, child	Investigational: trial-specific dosing is in the registry record	Not approved. Not yet recruiting, last update 2026-09-28	NCT07285798 . Primary outcome: Change From Baseline in the Aberrant Behavior Checklist Irritability (ABC-I) Score at Week 8. A registry entry does not show that the drug works.
KarX-EC	Emerging: investigational, Phase 3	Registered conditions: Irritability Associated With Autism Spectrum Disorder. Ages: from 5 Years, up to 17 Years, child	Investigational: trial-specific dosing is in the registry record	Not approved. Not yet recruiting, last update 2026-09-28	NCT07285798 . Primary outcome: Change From Baseline in the Aberrant Behavior Checklist Irritability (ABC-I) Score at Week 8. A registry entry does not show that the drug works.
Lumateperone	Emerging: investigational, Phase 3	Registered conditions: Schizophrenia; Bipolar Disorder; Autism Spectrum Disorder. Ages: from 5 Years, up to 17 Years, child	Investigational: trial-specific dosing is in the registry record	Not approved. Recruiting, last update 2026-09-25	NCT06229210 . Primary outcome: Incidence of Common Adverse Events. A registry entry does not show that the drug works.
Sertraline	Emerging: investigational, Phase 2	Registered conditions: Autism Spectrum Disorder; Anxiety. Ages: from 8 Years, up to 14 Years, child	Investigational: trial-specific dosing is in the registry record	Not approved. Completed, last update 2026-09-23	NCT03279471 . Primary outcome: Change in Pediatric Anxiety Rating Scale. A registry entry does not show that the drug works.

Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research. Food and Drug Administration today initiated the approval of leucovorin calcium tablets for patients with cerebral folate deficiency (CFD), a neurological condition that affects folate (a vitamin essential for brain health) transport into the brain.
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research. Food and Drug Administration today initiated the approval of leucovorin calcium tablets for patients with cerebral folate deficiency (CFD), a neurological condition that affects folate (a vitamin essential for brain health) transport into the brain.
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research. Food and Drug Administration today initiated the approval of leucovorin calcium tablets for patients with cerebral folate deficiency (CFD), a neurological condition that affects folate (a vitamin essential for brain health) transport into the brain.
date not stated	Regulatory announcement	FDA Takes Action to Make a Treatment Available for Autism Symptoms / FDA	"The FDA is collaborating with GSK to broaden the existing Wellcovorin label," said George Tidmarsh, M.D., Ph.D., Director of the FDA's Center for Drug Evaluation and Research. Food and Drug Administration today initiated the approval of leucovorin calcium tablets for patients with cerebral folate deficiency (CFD), a neurological condition that affects folate (a vitamin essential for brain health) transport into the brain.
date not stated	Regulatory announcement	[[PDF] guideline-clinical-development-medical-products-treatment-autism ...](https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-clinical-development-medical-products-treatment-autism-spectrum-disorder-asd_en.pdf)	No sentence stating the action was retrieved; open the page

Page wording from the regulator's own site; read the page for the full announcement.

Safety warnings (FDA boxed warnings of the drugs listed above)

Drug(s) (FDA label)	Boxed warning, in the label's words
Quetiapine , Lumateperone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death [see Warnings and Precautions (5.1)]."
Ziprasidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Ziprasidone is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
Paliperidone , Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Paliperidone extended-release tablets are not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Sertraline	"Suicidality and Antidepressant Drugs Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders. Anyone considering the use of Sertraline hydrochloride or any other antidepressant in a child, adolescent, or young adult must balance this risk with the clinical need."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."

These are the labels' own boxed warnings, not a complete safety review: read the full label.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT05307744	PS128; Placebo	Phase not applicable	Terminated (2026-10-06)	Terminated early (During COVID-19, study execution); results posted (not retrieved here).
NCT06659328	Structured Walking Intervention for Children with Autism Spectrum Disorder.	Phase not applicable	Completed (2026-10-06)	Completed; results not posted/retrieved.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NT1164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07243561	hUC-MSC-Exos Nasal Spray	Phase 1/2	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.

2 further registry record(s) about assessment, characterisation or peripheral research were retrieved and are not shown (they are not treatment trials).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, meta-analysis: "Exercise significantly improved overall, gross, and fine motor skills in children with ASD." [1]
2. **Hao et al. (2026)**, systematic review: "There was a significant increase in XR autism research after 2020, with virtual reality (VR) as the largest modality (n = 61, 53.0%), followed by augmented reality (AR) (n = 36, 31.3%)." [5]
3. **Mazurek et al. (2026)**, systematic review: "Most studies (95%) found that behavioral interventions were effective for at least some sleep outcomes." [2]
4. **Restoy et al. (2024)**, meta-analysis: "Studies of non-pharmacological interventions showed significant improvement in both ER and ED." [8]
5. **Sandbank et al. (2023)**, meta-analysis: "When effects were then restricted to exclude those at high risk of detection bias, only one significant summary effect was estimated-naturalistic developmental behavioral interventions on measures of diagnostic characteristics of autism (0.30, 0.03 to 0.57; P=0.03)." [9]

Conflicting evidence

None was retrieved.

What remains under investigation

- **Efficacy:** Head-to-head trials comparing exercise, behavioral sleep interventions, and ketogenic diet for long-term outcomes in ASD.
- **Effectiveness:** Real-world implementation of caregiver-assisted NDBI in diverse settings (e.g., community-based programs).
- **Feasibility:** Scalability of extended reality (XR) interventions for motor/social skills in clinical practice.
- **Association:** Longitudinal studies on sleep spindle characteristics as neurophysiological biomarkers for ASD progression.
- **Exploratory:** Dose-response relationships for music therapy on social communication (protocol-only; no results yet).

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 2 retrieved record(s) were excluded because they were not about the question's subject.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. [PMID 42758334](#). DOI 10.1007/s00431-026-07421-4.
2. Mazurek MO, Smith JV, Kiernan B, et al. Behavioral interventions for sleep problems in autistic children: A systematic review. *Sleep medicine*. 2026 Dec. [PMID 42685447](#). DOI 10.1016/j.sleep.2026.109242.
3. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. [PMID 42763398](#). DOI 10.1186/s12887-026-07708-3.
4. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. [PMID 42494078](#). DOI 10.1177/13623613261462574.
5. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
6. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
7. Hitomi K, Katayama S, Nakada S, et al. Crohn's Disease Presenting as an Abdominal Wall Abscess in an Adolescent With Autism Spectrum Disorder. *Case reports in pediatrics*. 2026-09-22. [PMID 42781187](#). DOI 10.1155/crpe/8225541.
8. Restoy D, Oriol-Escudé M, Alonzo-Castillo T, et al. Emotion regulation and emotion dysregulation in children and adolescents with Autism Spectrum Disorder: A meta-analysis of evaluation and intervention studies. *Clinical psychology review*. 2024 Apr. [PMID 38401510](#). DOI 10.1016/j.cpr.2024.102410.
9. Sandbank M, Bottema-Beutel K, Crowley LaPoint S, et al. Autism intervention meta-analysis of early childhood studies (Project AIM): updated systematic review and secondary analysis. *BMJ (Clinical research ed.)*. 2023 Nov 14. [PMID 37963634](#). DOI 10.1136/bmj-2023-076733.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)
- [FDA Takes Action to Make a Treatment Available for Autism Symptoms | FDA](#)

- [[PDF] guideline-clinical-development-medicinal-products-treatment-autism ...](https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-clinical-development-medicinal-products-treatment-autism-spectrum-disorder-asd_en.pdf)
 - [FDA drug label: Aripiprazole \(label effective 2026-03-05\)](#)
 - [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)
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Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.